Game Title

Quantum aeroplane chess (TBD)

Developer(s)

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Core Concept

Quantum aeroplane chess is a digital board game inspired by Aeroplane Chess where players control aircraft delivering aid to crisis zones across a global map. Instead of a single race to a finish point, players compete to reach multiple dynamic targets that appear during the game. Each crisis zone represents a location where aid is needed, and players earn points by successfully delivering assistance.

The gameplay keeps the simple and recognizable structure of a turn-based board game, but introduces a strategic layer through quantum-inspired mechanics such as superposition, measurement, and entanglement. Aircraft may temporarily exist in several possible movement outcomes before the player commits to a final destination. Players must balance risk and certainty while competing to gain the highest humanitarian impact score.

The design focuses on making a fun and strategic digital board game first, while integrating quantum-inspired mechanics that create unique gameplay situations and player choices.

Thematic Focus

The main theme of the game is humanitarian response to global crises, connected to social good through:

- SDG 13: Climate Action
- SDG 11: Sustainable Cities and Communities

Crisis zones appear across the world map representing disaster situations such as extreme weather events, infrastructure failures, or humanitarian emergencies. Players compete to respond to these crises with their aircraft.

The theme is integrated directly into gameplay mechanics:

- Crisis zones appear dynamically and increase in severity if ignored.
- Delivering aid to a crisis zone earns impact points.
- Some crises become more valuable but riskier over time.

The system creates competition between players while representing the growing pressure of unresolved global crises.

The game remains primarily competitive rather than cooperative, focusing on strategic decision-making and replayability.

Quantum Dimension

Quantum concepts used

- Superposition: aircraft movement can exist in multiple possible positions before being resolved.
- Measurement: players choose when to collapse the movement state into a final position.
- Collapse: once measured, the aircraft moves to a specific location.
- Entanglement: certain events can temporarily link aircraft so that actions affecting one may influence another.
- Interference: overlapping movement possibilities can amplify or cancel movement outcomes.

Level of implementation

Mechanical + state-based implementation.

The game does not simulate full quantum computation, but models quantum-like behaviors through internal game states such as:

- movement superposition states
- entanglement links between aircraft
- collapse events triggered by measurement
- interference effects when paths overlap

These systems introduce strategic timing and risk management rather than simple random movement.

How it goes beyond classical probabilistic randomness

A classical board game would typically use dice rolls to determine movement. In this design, outcomes are not immediately resolved.

- Movement possibilities exist simultaneously until measured.
- Players can delay measurement to maintain multiple potential outcomes.
- Entanglement links the behavior of different aircraft rather than resolving actions independently.
- Interference can amplify or cancel movement effects when paths overlap.

Because outcomes evolve before resolution, the system introduces strategic control over uncertainty rather than pure randomness.

Game Mechanics and Rules

Turn structure (turn-based)

Each round follows a simple sequence:

1. Player Phase
    - Player selects one aircraft and chooses an action.
2. Resolution Phase
    - Movement or quantum states are updated.
3. Crisis Update Phase
    - Crisis zones may appear or increase in severity.
4. End Phase
    - State timers and effects update.

Player actions and choices

Players control several aircraft and choose between different movement options:

- Direct Move: safe movement with predictable outcome.
- Quantum Move: creates multiple possible movement outcomes.
- Measure: collapse the aircraft's superposition into a final position.
- Entangle: temporarily link aircraft states (special event or ability).

Players must decide when to commit to movement and which crisis zones to prioritize.

System responses

The game map evolves during play:

- New crisis zones appear at random locations.
- Existing crises increase in severity if ignored.
- Higher severity zones provide higher rewards.

A global instability meter increases if crises remain unresolved for too long.

Constraints and limitations

- Limited number of aircraft per player.
- One primary action per turn.
- Global instability meter that increases over time.
- Quantum movement introduces risk if collapse occurs in an unfavorable position.

Narrative / Setting

The game takes place in a near-future world where global crises appear frequently due to environmental and social instability. Players represent competing humanitarian aviation teams responding to emergencies around the world.

The narrative serves mainly as a thematic framework for gameplay. The focus remains on dynamic map events, strategic movement decisions, and competition between players.

Crisis zones and aircraft missions symbolize the challenge of responding to global emergencies in an uncertain environment.

Winning and Losing Conditions

Winning (game-level)

The winner is the player who achieves the highest humanitarian impact score by delivering aid to crisis zones.

Points are earned by:

- Successfully reaching crisis zones.
- Resolving high-severity crises.
- Efficient movement and risk management.

Losing (game-level)

The game ends in a shared loss if:

- The global instability meter reaches its maximum value.
- Too many crisis zones escalate beyond control.

This condition prevents players from ignoring the global system entirely.

Platform and Tools

- Engine: Godot (2D)
- Language: GDScript
- Assets: 2D sprites, board-style map, aircraft icons, simple UI panels

Planned technical scope

Playable prototype including:

- 2–4 players
- Global board map with multiple crisis zones
- Quantum movement system (superposition + collapse)
- Entanglement and interference mechanics
- Instability meter system

- Turn-based gameplay

The goal is to develop a polished prototype demonstrating the core gameplay systems.